

Figures Theoretical Mechanics

Felix Claeys, Brecht Verbeken, Simon Verbruggen, Jorrit Vander Bracht

March 26, 2026

1.2 Mechanics of Multiple Particles

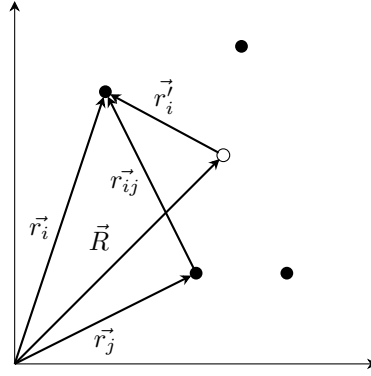


Figure 1: Vector notations for points of mass and the center of mass.

1.3 Bindings

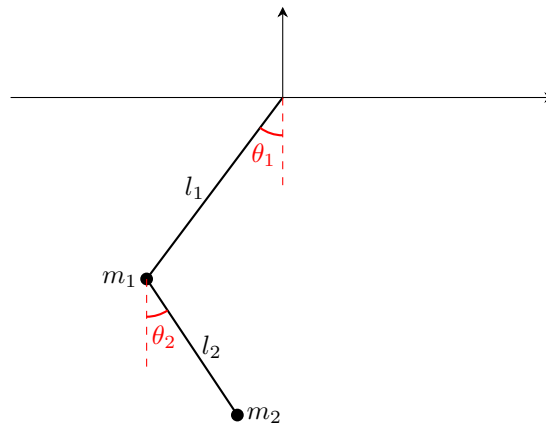


Figure 2: Double pendulum.

1.4 lagrange multipliers

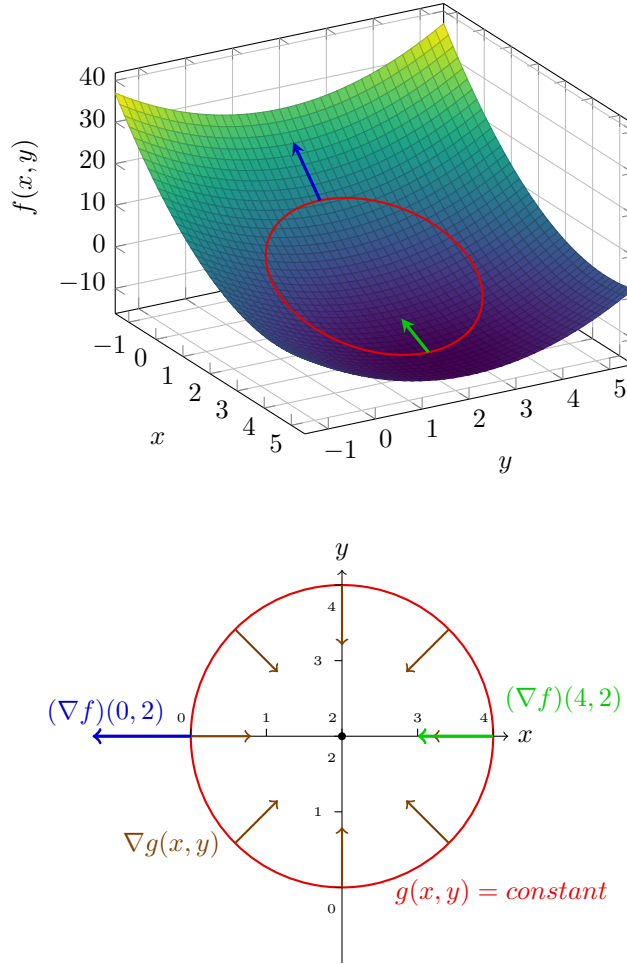


Figure 3: We seek to find the maxima and minima of $f(x, y) = (x - 3)^2 + (y - 2)^2 - 3x$ subject to the constraint $(x - 2)^2 + (y - 2)^2 = 4$, i.e., $g(x, y) = 4$. This constraint circle is shown in red on the surface $(x, y, f(x, y))$ in Figure 1. Maxima on this restricted curve do not require $\nabla f = 0$. Indeed, the non-zero gradient vectors in the extrema are drawn here, tangent to the surface along the curve (blue and green vectors). In Figure 2 (xy -plane), notice that only at extrema is ∇f parallel to ∇g . The gradient ∇g is always perpendicular to the level curve $g(x, y) = 4$ (brown arrows). Thus, a sufficient condition for extrema is $\nabla f = \lambda \nabla g$, where λ are the Lagrange multipliers. This gives, together with the equation $g(x, y) = 4$, a system that can be solved for the extrema.

1.6 Atwoods Machine

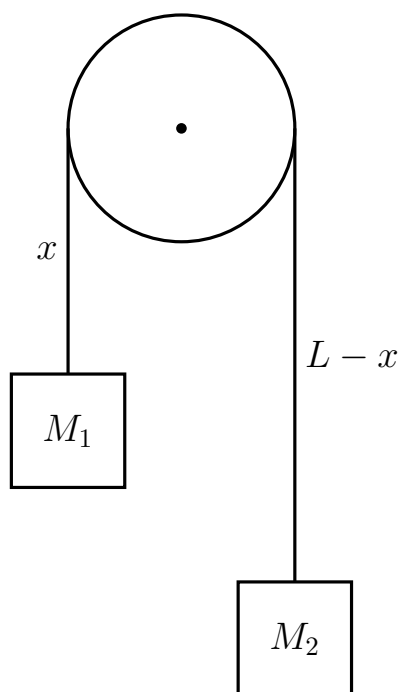


Figure 4: Atwood's machine.

2.6 Conservation Properties and Symmetry Properties

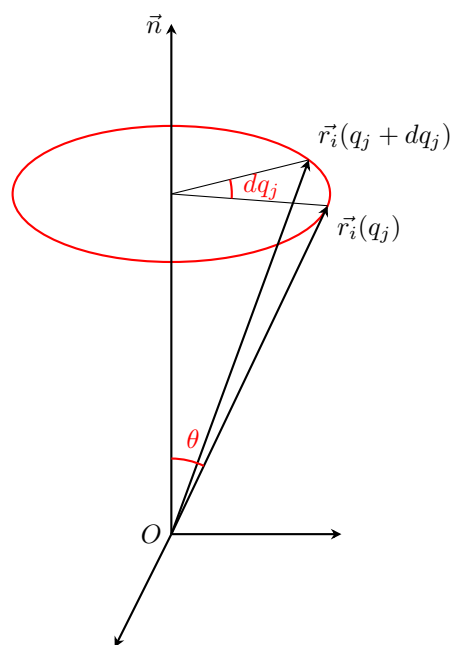


Figure 5: Change of position vector under rotation.

4.4 Euler angles

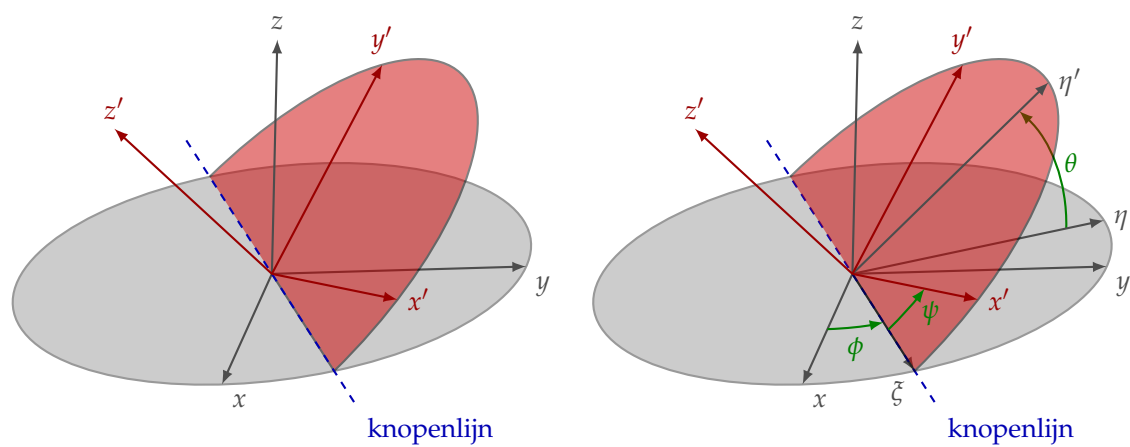


Figure 6: The rotations defining the Eulerian angles.

4.7 Finite Rotations Detail

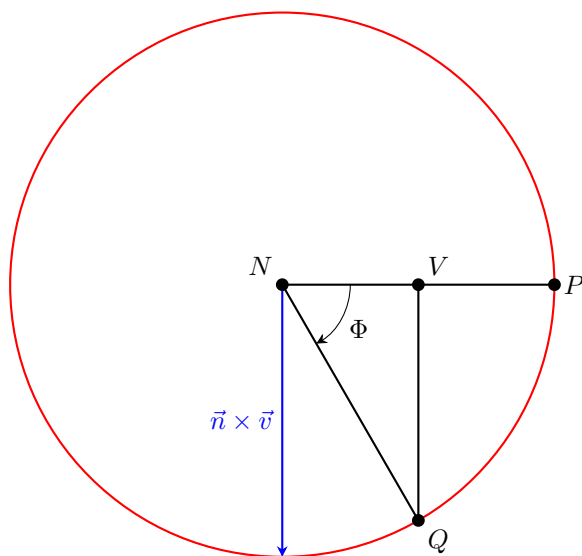


Figure 7: Detail image of the circle on which the rotation takes place.

4.7 Finite Rotations

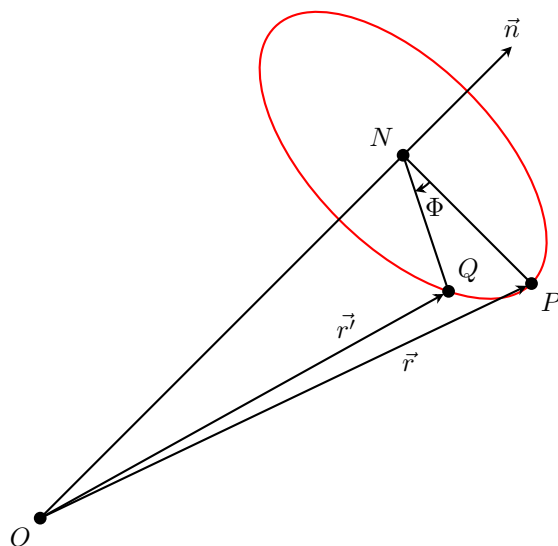


Figure 8: Finite rotation around an axis.

4.8 Infinitesimal rotations

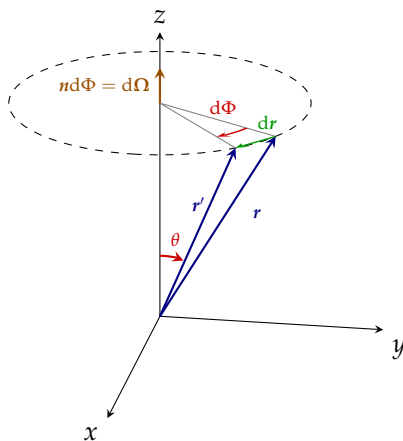


Figure 9: Change in a vector $\mathbf{r}$ produced by an infinitesimal clockwise rotation $d\Phi$ of the vector.

4.9 Rate of change of a vector

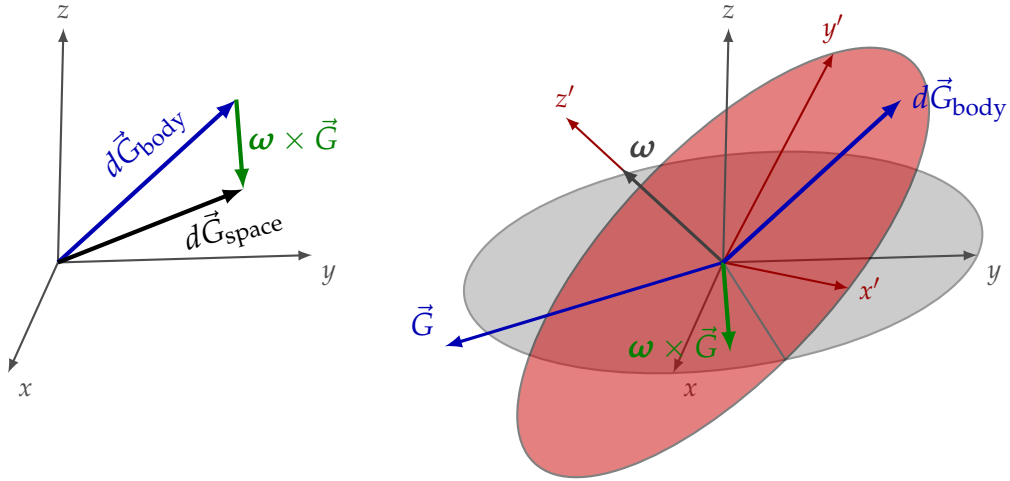


Figure 10: Rate change of a vector $\vec{G}$, as perceived by an observer in the fixed body frame: $d\vec{G}_{body}$ (right) and in the space frame: $d\vec{G}_{space} = d\vec{G}_{body} + \vec{\omega} \times \vec{G}$ (left). The z' axis is chosen such that it aligns with $\vec{\omega}$.

4.10.1 coriolis on sphere

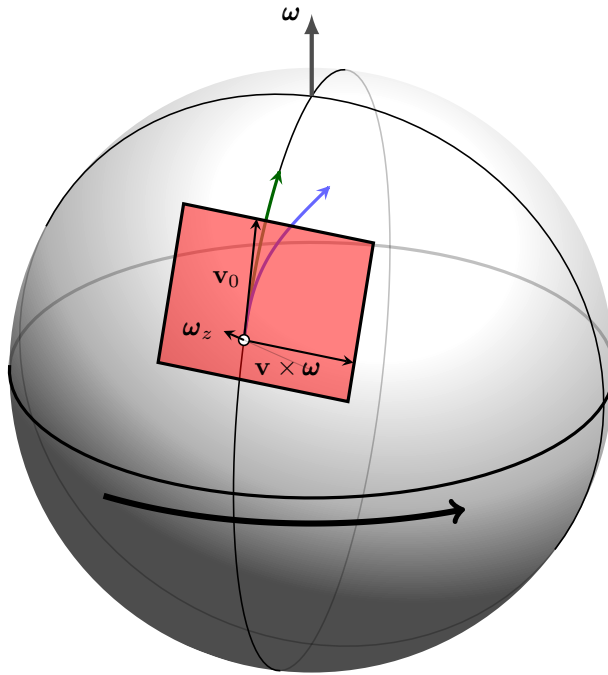


Figure 11: Coriolis effect for an object moving northward on a rotating sphere. The green trajectory shows the motion for when the Coriolis effect is neglected, while the blue trajectory takes the Coriolis effect into account.

4.10.2 coriolis effect local

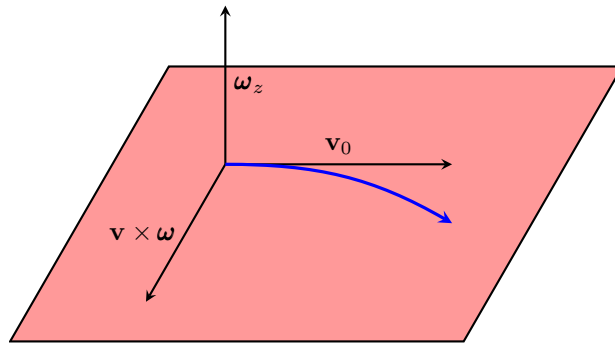


Figure 12: Local view of Coriolis deflection (blue) for an object moving northward.

4.10.3 centrifugal force

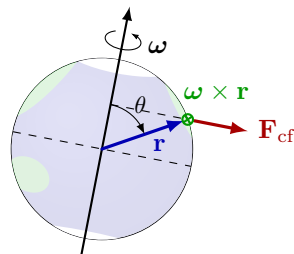


Figure 13:

4.10.3 coriolis sphere sol

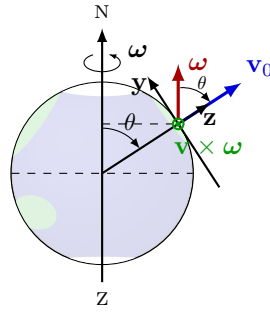


Figure 14:

4.10.3 coriolis sphere task

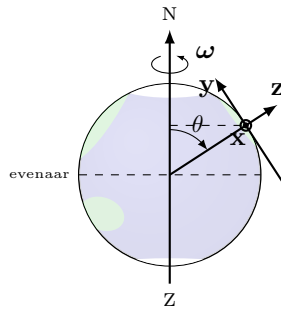


Figure 15:

4.10.4 coriolis local task

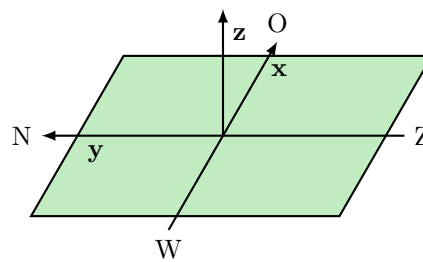


Figure 16:

4.10.4 coriolis local sol

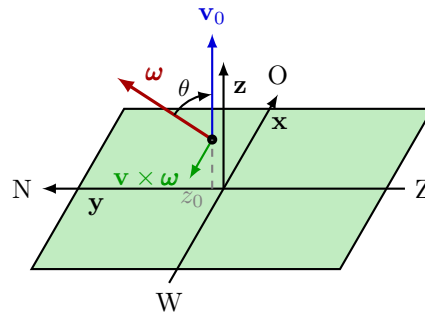


Figure 17:

4.10.4.1 coriolis forces northern hemisphere

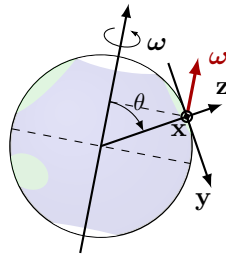


Figure 18:

4.10.4.2 coriolis forces northern hemisphere

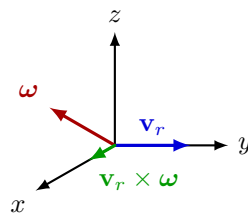


Figure 19:

4.10.5.1 coriolis forces southern hemisphere

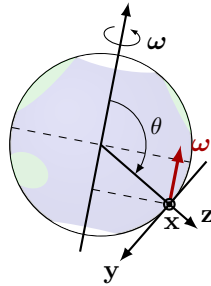


Figure 20:

4.10.5.2 coriolis forces southern hemisphere

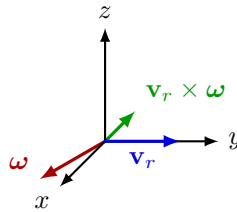


Figure 21:

4.10.6 coriolis effect vortex

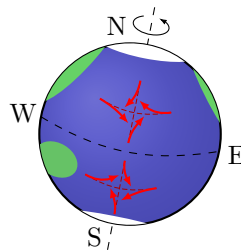


Figure 22:

4.10.7.1 coriolis NH vertical velocity

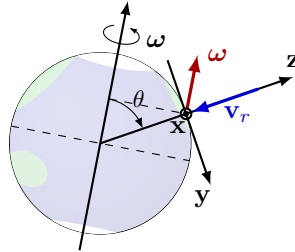


Figure 23:

4.10.7.2 coriolis NH vertical velocity

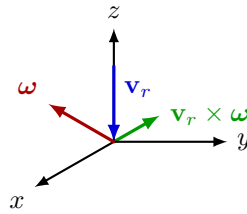


Figure 24:

4.10.8.1 coriolis SH vertical velocity

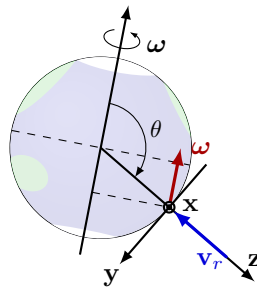


Figure 25:

4.10.8.2 coriolis SH vertical velocity

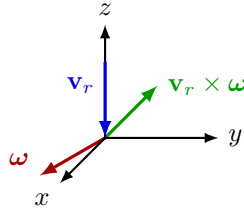


Figure 26:

5.2 traagheidstensor balk 2D visualisatie

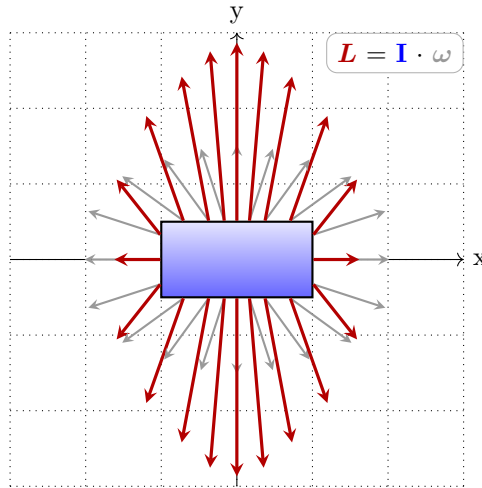


Figure 27: Graphical representation of the relationship between the angular velocity vector ω (gray) and the angular momentum vector $\mathbf{L}$ (red) for a rectangular beam with inertia tensor $\mathbf{I}$. The directions of $\mathbf{L}$ and ω do not coincide due to the anisotropy of the inertia tensor. The rectangular beam has dimensions 2, 1, and 0.2 along the x, y, and z axes respectively and is homogeneous in density.

5.3 visual proof parallel axis theorem for point-symmetrical bodies

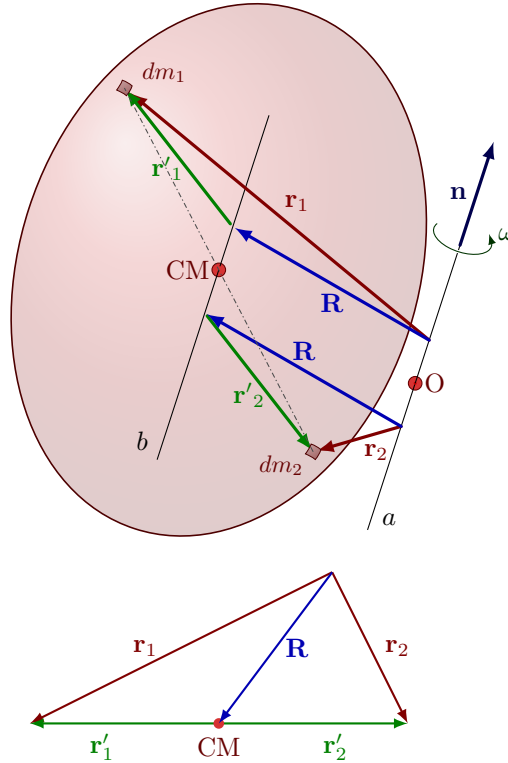


Figure 28: One wants to prove that $I_a = I_b + M(\vec{R} \times \vec{n})^2$. For this, one can consider the 3D figure. Notice that all vectors in green, blue and red are perpendicular to the vector $\vec{n}$. For a point-symmetrical, homogeneous body, it is enough to show that for two point masses dm_1, dm_2 , $dI_a = dI_b + dI_{CM}$. Here, $dI_a = dm_2 r_2^2 + dm_1 r_1^2 = dm_1(r_1^2 + r_2^2)$, $dI_{CM} = 2dm_1 R^2$, and $dI_b = 2dm_1 r_1'^2$, as due to homogeneity $dm_1 = dm_2$. We then have $dI_{CM} + dI_b = 2dm_1(R^2 + r_1'^2)$, so for equivalence to hold one needs that $r_1^2 + r_2^2 = 2(R^2 + r_1'^2)$. This can be interpreted in the 2D triangle of the second figure. One finds this triangle by combining the two smaller triangles in the 3D figure and projecting them on a plane perpendicular to the vector $\vec{n}$. Notice that, as masses dm_1 and dm_2 were chosen symmetrically with respect to the CM, one has $r_1' = r_2'$. The relation $r_1^2 + r_2^2 = 2(R^2 + r_1'^2)$ then trivially holds due to Apollonius's theorem.

5.3 parallelaxistheorem

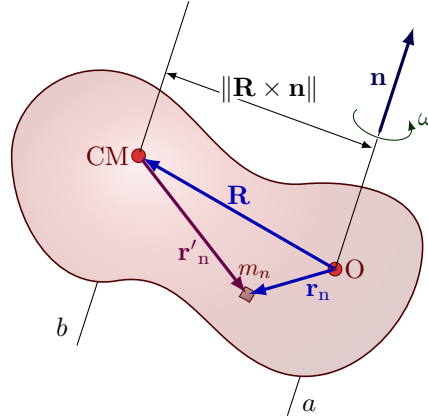


Figure 29:

5.3 traagheidstensor ellipsoïde 2D visualisatie

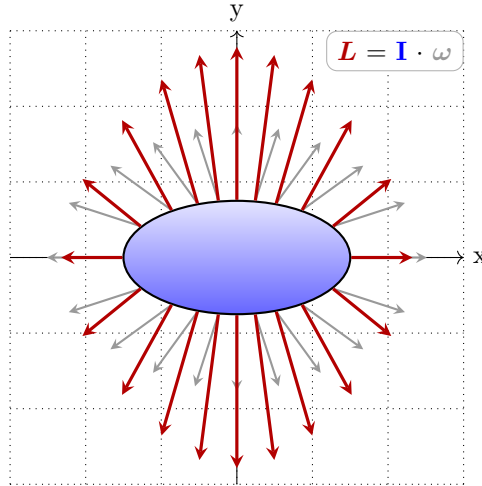


Figure 30: Graphical representation of the relationship between the angular velocity vector ω (gray) and the angular momentum vector $\mathbf{L}$ (red) for an ellipsoid with inertia tensor $\mathbf{I}$. The directions of $\mathbf{L}$ and ω do not coincide due to the anisotropy of the inertia tensor. The ellipsoid has dimensions 3, 1.5 and 1.5 along the x, y, and z axes respectively and is homogeneous in density.

5.7 Heavy symmetrical top with one point fixed

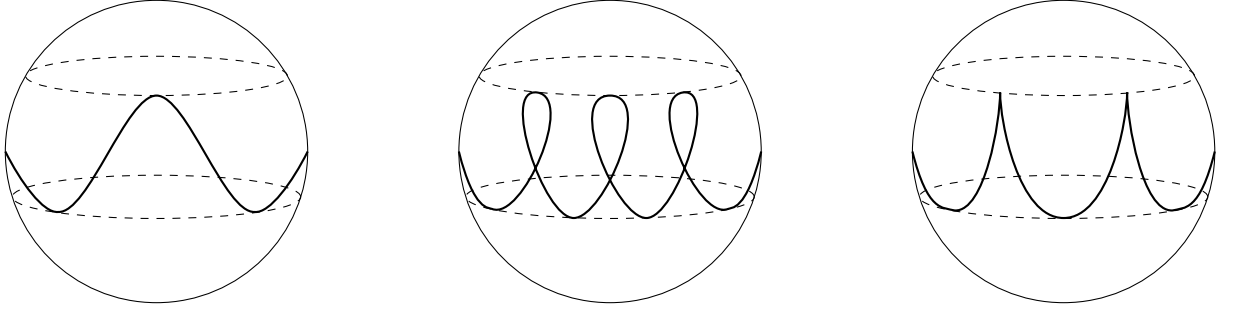


Figure 31: From left to right, monotone precession, spin-reversal precession and non-monotone precession.

6.2 water

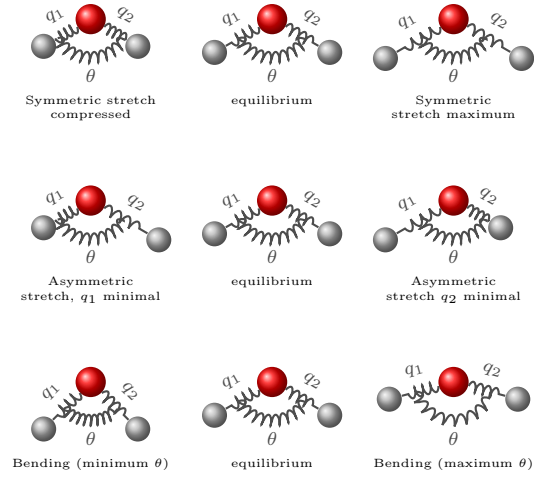


Figure 32: Vibrational modes of H_2O : $3N=9$ degrees of freedom total. 3 vibrational (shown), 3 translational, 3 rotational. Modes: symmetric stretch ($q_1 = q_2$), asymmetric stretch ($q_1 \neq q_2$), bending (θ).

6.2 carbondioxide

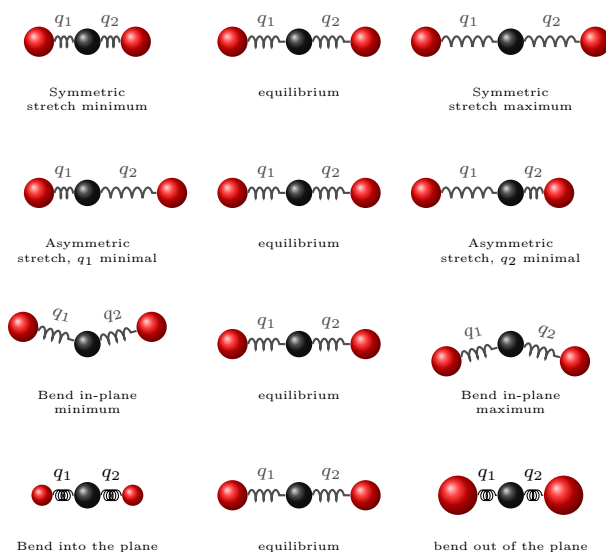


Figure 33: Vibrational modes of CO_2 : $3N=9$ degrees of freedom total. 4 vibrational (shown: symmetric and asymmetric stretch, bending in/out-plane), 3 translational, 2 rotational. Bending doubly degenerate (linear molecule). The springs between the C-O bonds and the corresponding generalized coordinate θ are omitted to avoid overloading the figure.

6.3 Normal Coordinates

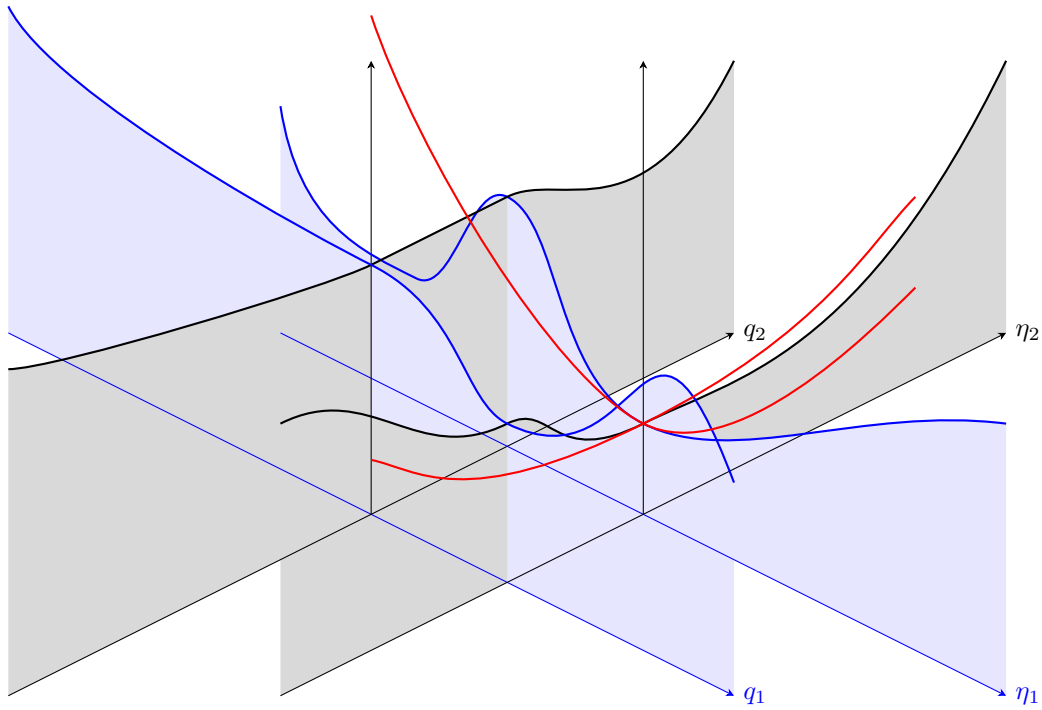


Figure 34: Expansion around a small diversion in a minimum up to second order.

8.1 Afbeelding1

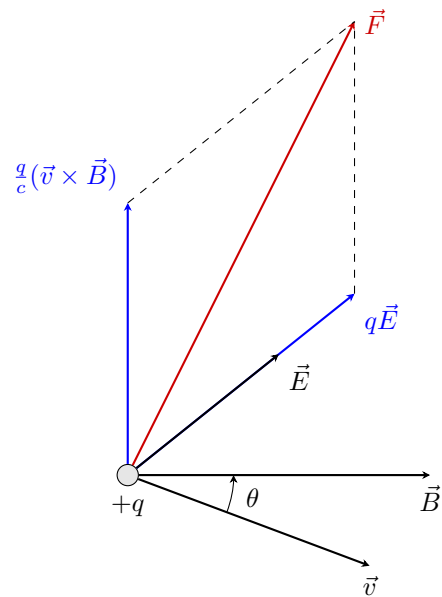


Figure 35:

8.1 Slide 187b

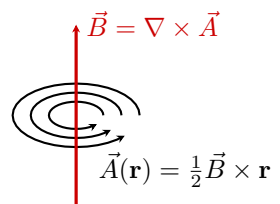


Figure 36:

8.1 Slide 187

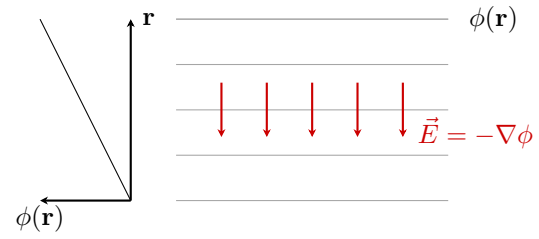


Figure 37: